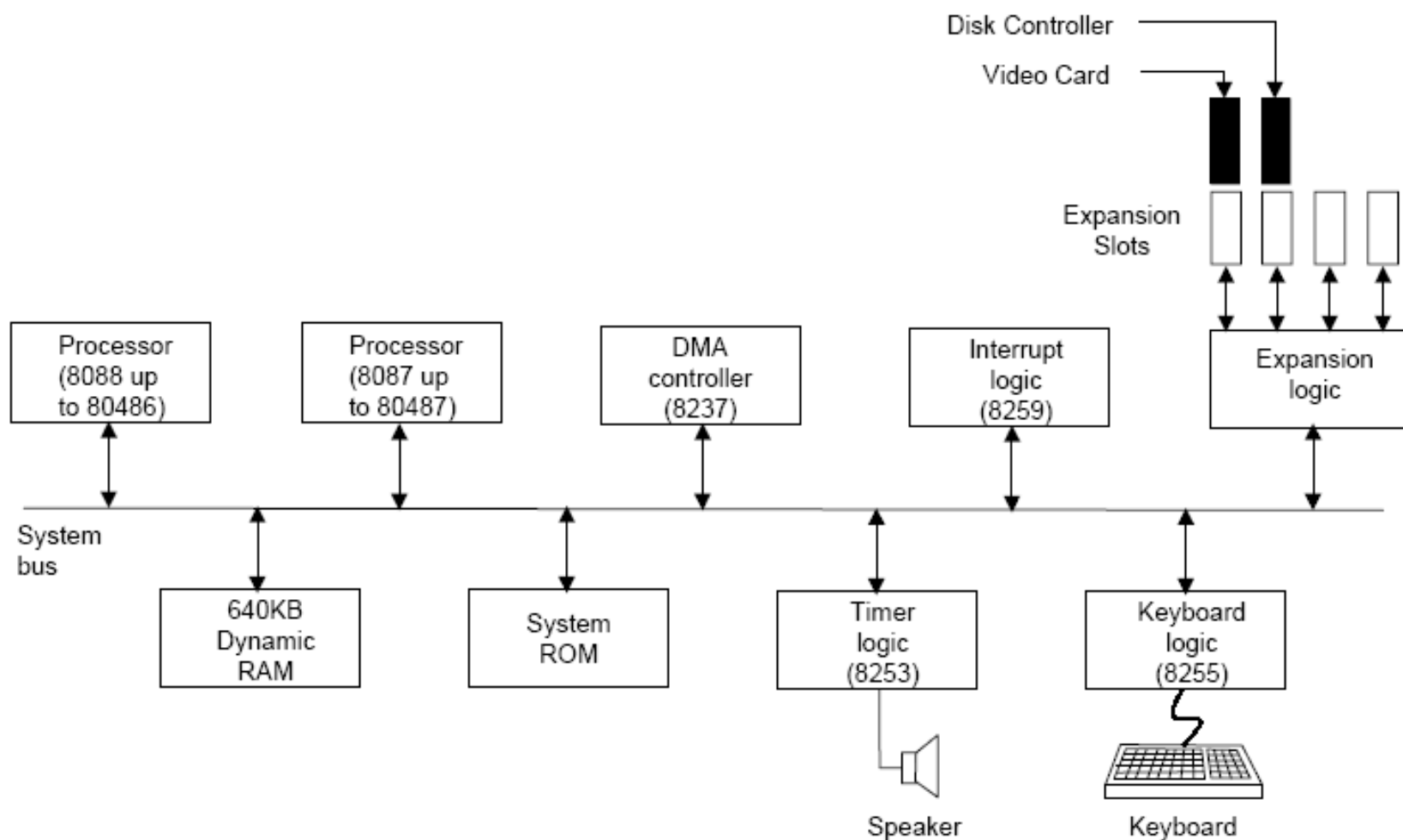


Arhitektura račūnala

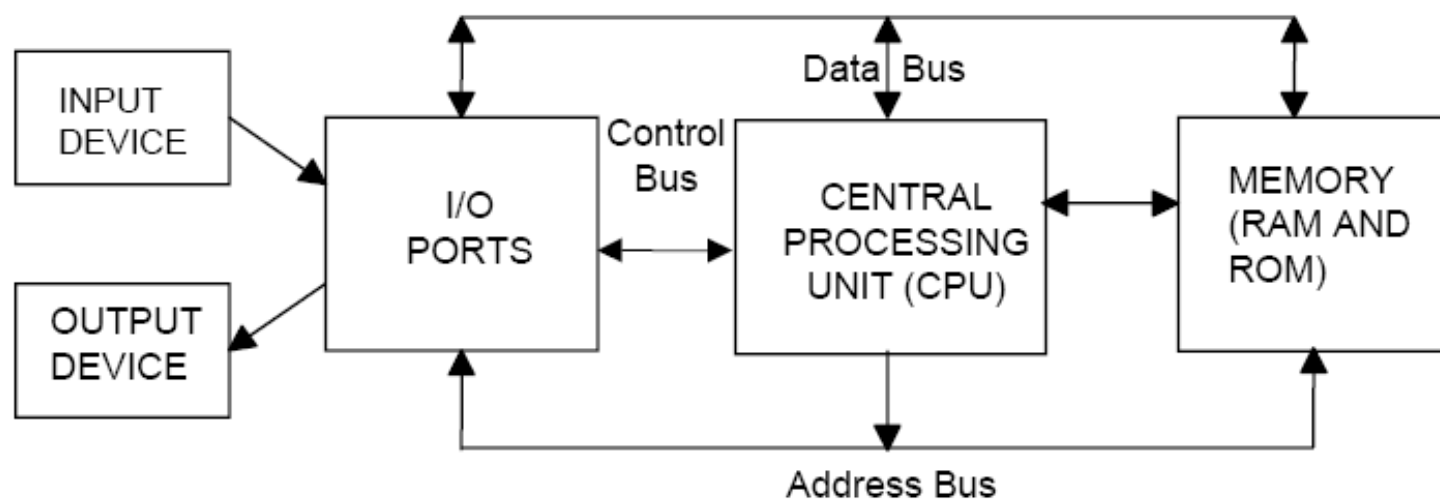
LV 2

- Ponavljanje
 - Vrste podataka...
 - Upis podataka u računalu/memoriji
-
- Slijedi
 - ATMEL Studio
 - Arhitektura računala
 - Memorija
 - Upis podataka u računalu/memoriji

Klasična matična ploča



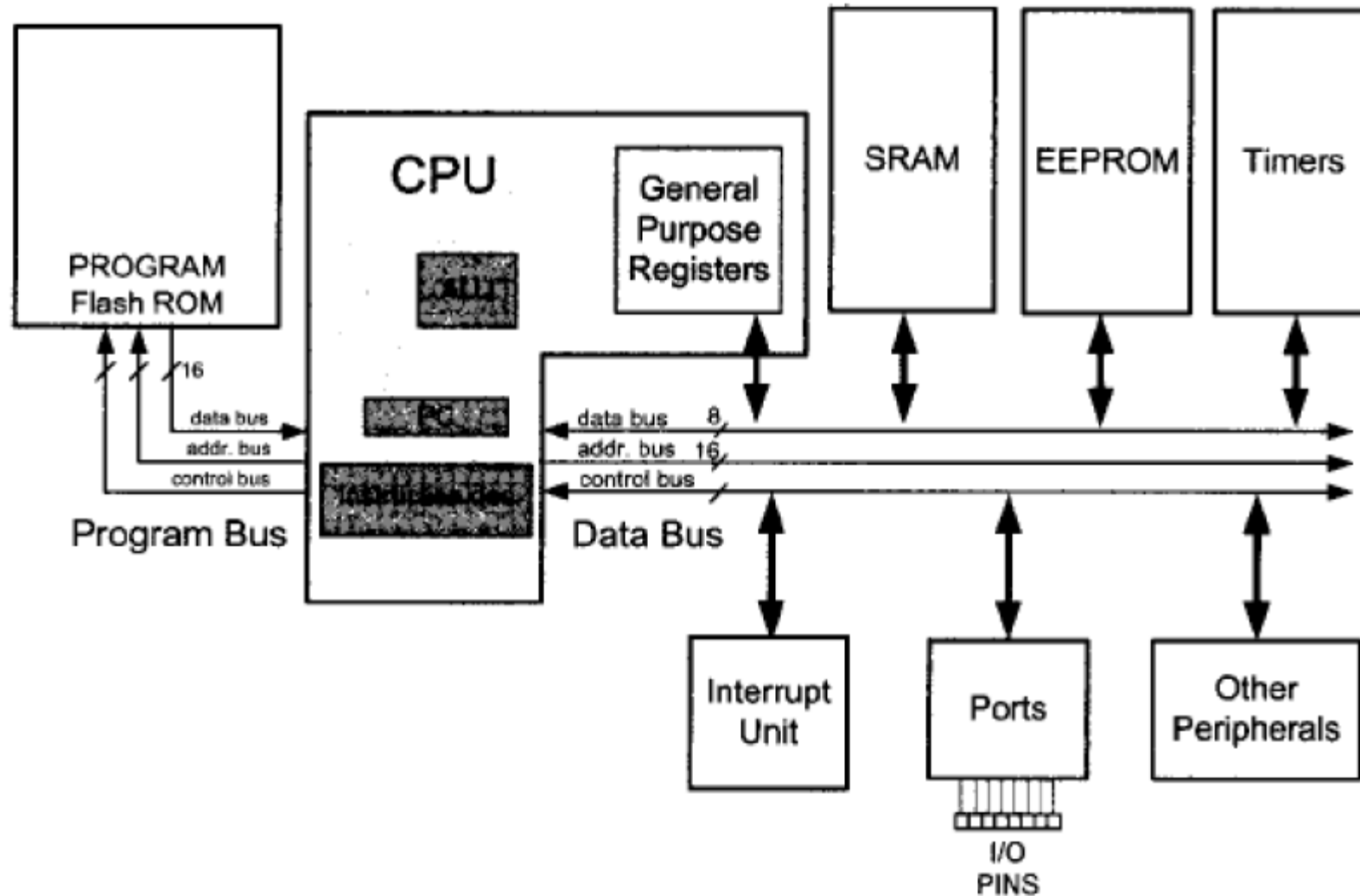
Mikroupravljač



Major parts: *CPU, Memory, I/O*

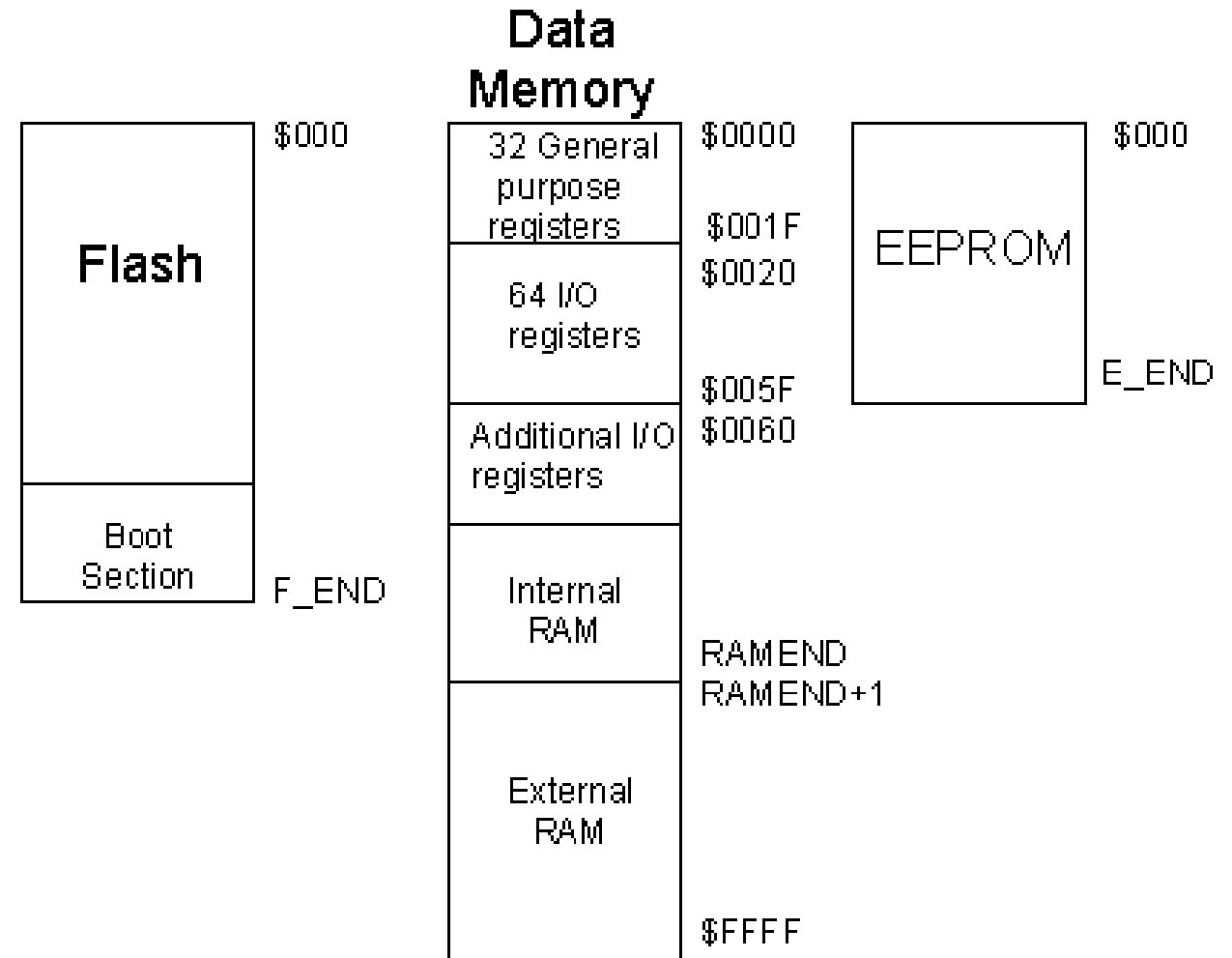
Buses: *address bus, data bus, and control bus.*

AVR mikroupravljač - Harvard architektura



Pogramaska memorija ATMEGA 328p vs. 2560

- ATMEGA 328p vs. 2560
- **Flash**
- 0x0000–0x1FFF/ 0x3FFF
- Program Counter (PC) -14/ 17bits
- Program memory is addressable on words(2bytes): 16K x 16 biti(32KB) / 128Kx16 bits (256 KB)
- Program memory cannot be extended



Podatkovna memorija ATMEGA 328p vs. 2560

Data memory:

- is addressable on bytes
- Registers: 32 GPR, 64 I/O, 160/416 extended I/O
- (0x0000 –0x001F): General purpose registers (GPR):
- (0x0020 –0x005F): direct I/O access or as memory
- (0x0060 –0x00FF / 0x01FF): Extended I/O in SRAM, access with ST/STS/STD si LD/LDS/LDD
- (0x0100 –0x08FF / 0x0200 –0x21FF): internal SRAM

Podatkovna memorija ATMEGA 328p vs. 2560

- ATMEGA 328p

32 Registers	0x0000 - 0x001F
64 I/O Registers	0x0020 - 0x005F
160 Ext I/O Reg.	0x0060 - 0x00FF
Internal SRAM 2048 x 8	0x0100 0x08FF

- ATMEGA 2560

Address (HEX)	
0 - 1F	32 Registers
20 - 5F	64 I/O Registers
60 - 1FF	416 External I/O Registers
200	Internal SRAM (8192 x 8)
21FF	
2200	External SRAM (0 - 64K x 8)
FFFF	

Uvod u Atmel studio 7.0

- Atmel studio...

Instrukcije

- Pohranjene u programskoj memoriji
- Instrukcija se sastoji od dva dijela:
 1. Operacija koja se izvodi – operacijski kod
 2. Odakle dolaze operandi i kuda se sprema rezultat operacije - adrese
- Instrukcije pohranjene u memoriji → “nizovi bitova”
- Svakom nizu bita dodjeljeno odabrano značenje → kodiranje
- Kodu koji se sastoji od “0” i “1” pridodaje se slovčana kratica, a rezultatu i operandu simbolička imena

Zapis instrukcije u memoriji

- Primjer programa 1.

start:

inc R1

jmp start

- Izgled memorije →

The screenshot shows a software window titled "AssemblerApplication4" with a tab for "main.asm". The assembly code is as follows:

```
; Replace with your application code
start:
inc R1
jmp start
```

Below the code editor is a "Memory 3" panel. It shows a memory dump for "prog FLASH" starting at address "0x0000,prog". The dump consists of 16 rows, each representing a 16-byte memory location. The first row (0x0000) contains the values 13, 94, 0c, 94, followed by 12 zeros. All other rows contain 16 zeros.

Memory:	prog FLASH	Address:	0x0000,prog
prog 0x0000	13 94 0c 94 00 00 ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff		
prog 0x0016	ff ff		
prog 0x002C	ff ff		
prog 0x0042	ff ff		
prog 0x0058	ff ff		
prog 0x006E	ff ff		
prog 0x0084	ff ff		
prog 0x009A	ff ff		
prog 0x00B0	ff ff		
prog 0x00C6	ff ff		
prog 0x00DC	ff ff		

At the bottom of the memory panel, there are tabs for "Memory 3", "Autos", "Locals", "Watch 1", and "Watch 2".

73. LDI – Load Immediate

73.1. Description

Loads an 8-bit constant directly to register 16 to 31.

Operation:

(i) $Rd \leftarrow K$

Syntax:

(i) LDI Rd,K

Operands:

$16 \leq d \leq 31, 0 \leq K \leq 255$

Program Counter:

$PC \leftarrow PC + 1$

16-bit Opcode:

1110	KKKK	dddd	KKKK
------	------	------	------

65. INC – Increment

65.1. Description

Adds one -1- to the contents of register Rd and places the result in the destination register Rd.

The C Flag in SREG is not affected by the operation, thus allowing the INC instruction to be used on a loop counter in multiple-precision computations.

When operating on unsigned numbers, only BREQ and BRNE branches can be expected to perform consistently. When operating on two's complement values, all signed branches are available.

Operation:

$$(i) \quad Rd \leftarrow Rd + 1$$

Syntax:

Operands:

Program Counter:

$$(i) \quad \text{INC Rd}$$

$$0 \leq d \leq 31$$

$$PC \leftarrow PC + 1$$

16-bit Opcode:

1001	010d	dddd	0011
------	------	------	------

66. JMP – Jump

66.1. Description

Jump to an address within the entire 4M (words) Program memory. See also RJMP.

This instruction is not available in all devices. Refer to the device specific instruction set summary.

Operation:

(i) $PC \leftarrow k$

Syntax:

Operands:

Program Counter:

Stack:

(i) JMP k

$0 \leq k < 4M$

$PC \leftarrow k$

Unchanged

32-bit Opcode:

1001	010k	kkkk	110k
kkkk	kkkk	kkkk	kkkk

85. NOP – No Operation

85.1. Description

This instruction performs a single cycle No Operation.

Operation:

- (i) No

Syntax:

Operands:

Program Counter:

- (i) NOP

None

$PC \leftarrow PC + 1$

16-bit Opcode:

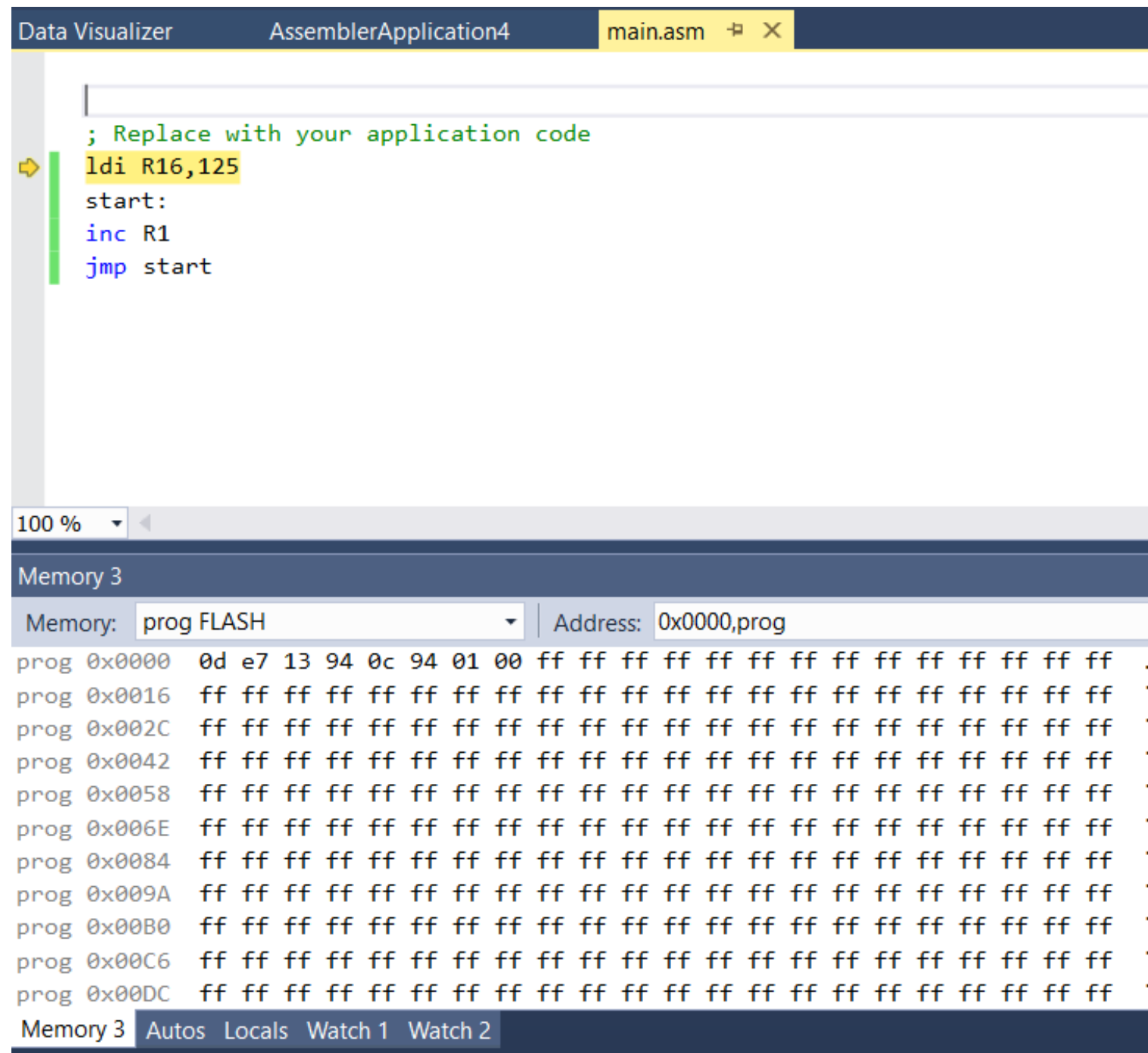
0000	0000	0000	0000
------	------	------	------

- Primjer programa 2.

start:

```
jmp start
```

- Izgled memorije →



Zadatak

- Odredi izgled programske memorije nakon upisa sljedećeg programa

ldi R1,125

nop

start:

inc r16

nop

jmp start

jmp kraj

kraj:

Rješenje

[illegible][illegible]

Naredbe s
podacima

- MOV

MOV – Copy Register

Description

This instruction makes a copy of one register into another. The source register R_r is left unchanged, while the destination register R_d is loaded with a copy of R_r .

Operation:

(i) $R_d \leftarrow R_r$

Syntax:

(i) MOV R_d, R_r

Operands:

$0 \leq d \leq 31, 0 \leq r \leq 31$

Program Counter:

$PC \leftarrow PC + 1$

16-bit Opcode:

0010	11rd	dddd	rrrr
------	------	------	------

Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
–	–	–	–	–	–	–	–

Naredbe s podacima

- LDI

LDI – Load Immediate

Description

Loads an 8-bit constant directly to register 16 to 31.

Operation:

(i) $Rd \leftarrow K$

Syntax:

(i) LDI Rd,K

Operands:

$16 \leq d \leq 31, 0 \leq K \leq 255$

Program Counter:

$PC \leftarrow PC + 1$

16-bit Opcode:

1110	KKKK	dddd	KKKK
------	------	------	------

Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
–	–	–	–	–	–	–	–

STS – Store Direct to Data Space

Naredbe s
podacima

- STS

Description

Stores one byte from a Register to the data space. For parts with SRAM, the data space consists of the Register File, I/O memory, and internal SRAM (and external SRAM if applicable). For parts without SRAM, the data space consists of the Register File only. The EEPROM has a separate address space.

A 16-bit address must be supplied. Memory access is limited to the current data segment of 64KB. The STS instruction uses the RAMPD Register to access memory above 64KB. To access another data segment in devices with more than 64KB data space, the RAMPD in register in the I/O area has to be changed.

This instruction is not available in all devices. Refer to the device specific instruction set summary.

Operation:

(i) $(k) \leftarrow Rr$

Syntax:

Operands:

Program Counter:

(i) STS k,Rr

$0 \leq r \leq 31, 0 \leq k \leq 65535$

$PC \leftarrow PC + 2$

32-bit Opcode:

1001	001d	dddd	0000
kkkk	kkkk	kkkk	kkkk

Status Register (SREG) and Boolean Formula

I	T	H	S	V	N	Z	C
–	–	–	–	–	–	–	–

Naredbe s podacima

- LD

The result of these combinations is undefined:

LD r28, Y+

LD r29, Y+

LD r28, -Y

LD r29, -Y

Using the Y-pointer:

Operation:		Comment:
(i) $Rd \leftarrow (Y)$		Y: Unchanged
(ii) $Rd \leftarrow (Y), Y \leftarrow Y + 1$		Y: Post incremented
(iii) $Y \leftarrow Y - 1, Rd \leftarrow (Y)$		Y: Pre decremented
(iv) $Rd \leftarrow (Y+q)$		Y: Unchanged, q: Displacement
Syntax:	Operands:	Program Counter:
(i) LD Rd, Y	$0 \leq d \leq 31$	$PC \leftarrow PC + 1$
(ii) LD Rd, Y+	$0 \leq d \leq 31$	$PC \leftarrow PC + 1$
(iii) LD Rd, -Y	$0 \leq d \leq 31$	$PC \leftarrow PC + 1$
(iv) LDD Rd, Y+q	$0 \leq d \leq 31, 0 \leq q \leq 63$	$PC \leftarrow PC + 1$

Zadatak 1

- Koristeći naredbu LDI spremi sljedeće podatke u zadane registre

Registar	Podatak
R16	$150_{(10)}$
R17	$-33_{(10)}$
R18	$AC_{(16)}$
R19	$01011110_{(2)}$
R20	$1_{(10)}$

Zadatak 2.

- Koristeći naredbe LDI, LD, MOV i STS realiziraj sljedeći zadatak.
 1. Spremi podatak $FA_{(16)}$ u registar R19
 2. Isti podatak premjesti (kopiraj iz registra) u registar R21
 3. Zatim spremi podatak $01010010_{(16)}$ u registar R17
 4. Sadržaj registra R17 premjesti na adresu 0017H u memoriji